Lecture 5

Building a Sustainable Genomic Surveillance Programme

Day 2 · 15:15–16:00 · 45 min
≈ 25 min talk + 20 min discussion

LO6: design a sustainable, outsourced genomic surveillance workflow for your own setting

What we can do on Monday

Collect → **Store** → **Send** → **Analyse** → **Act**

No sequencer is needed to start. We need:

- a **plan** and a clear question
- **stored isolates** with good **metadata**
- a sequencing **partner**
- **people** who can analyse the data

Choose a question first

Type	Example	What to sequence
Outbreak (reactive)	“Is this ward cluster one strain?”	All cluster isolates + a few unrelated controls
Routine (proactive)	“Which ESBL/ carbapenemase genes cause our SSIs?”	e.g. all resistant SSI isolates for 6 months
Research	“Are post-caesarean SSI *E. coli* from the community?”	A designed sample with matched community isolates

Start small.

50–100 well-chosen isolates with complete metadata beat 1,000 random ones.

Sampling and biobanking

- Store **every** SSI isolate with resistance of interest (at least ESBL, carbapenem-resistant, MRSA)
- –80 °C freezer or glycerol stocks; a **backup** at a second site if possible
- Keep an **isolate register** (spreadsheet): study ID, organism, date, ward, specimen, AST, freezer location
- Agree on labelling **before** you start

Metadata and data management

- Use a template aligned with **PHA4GE** fields
- **Coded IDs only.** Keep the ID → patient key secure and separate
- Name files consistently: `UGHE-SSI-0001_R1.fastq.gz`
- Back up raw reads in **two places**: they can't be replaced
- Galaxy histories are a free, reproducible record of every analysis

Ethics, agreements and data sharing

- **Surveillance and IPC:** usually covered by routine public-health practice
- **Research:** needs ethics approval (Rwanda National Ethics Committee + institutional IRB)
- **Sending isolates abroad:** material transfer agreement (MTA)
- **Data-sharing agreement:** who owns the data and how it is published
- **Share genomes publicly** (ENA / NCBI) with non-identifying metadata

KEY MESSAGE

Equity: local investigators should lead the analysis and authorship.

Partners in the region

RBC
Rwanda Biomedical Centre / National Reference Laboratory

Africa CDC PGI
Regional sequencing and training network

ASLM
AMR surveillance programmes such as MAAP

Other partners
Universities, NGOs, Fleming Fund projects, commercial services

Ask the room: which of these do we already work with?

Reporting results so they change practice

Audience	What they need
Clinicians	“This isolate carries an NDM carbapenemase. Avoid carbapenems; consult ID.”
IPC committee	Cluster summary, tree, timeline, recommended actions
Ministry / RBC	Quarterly summary of resistance genes and clones

> Use simple visuals (Pathogenwatch) and a one-page template. **Close the loop:** did the action reduce SSIs?

People: keeping the skills alive

- **2–3 “genomics champions”** per department
- A monthly **data clinic / journal club** for course graduates
- Self-paced learning: Galaxy Training Network, The Carpentries, Africa CDC, Wellcome Connecting Science
- Build genomics into **student teaching** (MGHD, MBBS research projects)

Funding

- Build sequencing costs into grants (AMR, surgery, maternal health)
- Funders active in Africa: Fleming Fund, Wellcome, Gates Foundation, Africa CDC, NIH / Fogarty, EDCTP

BUDGET PER ISOLATE

Sequencing + shipping + staff time + data storage

Your one-page action plan

1. **Question:** what do I want to know?
2. **Isolates:** which, how many, from where, when? Already stored?
3. **Metadata:** which fields? Who collects them?
4. **Partner:** who? Which platform? Cost? Timeline?
5. **Analysis:** which tools? Who does it?
6. **Ethics:** IRB? MTA? Data sharing?
7. **Communication:** who gets results, in what format?
8. **First step next week:** _____

Discussion

20 min · small groups, then the whole room

1. What is the **single biggest barrier** to starting genomic surveillance at our institution? How could we overcome it?
2. Which **question** (outbreak, routine, research) would be most valuable to answer first?